

Ethereum L1 Strawmap 研究报告

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研究对象：<https://strawmap.org/> 与其 Google Drawing 导出、所有可访问超链接、EIP/Forkcast/ethresear.ch/GitHub/EF 公开资料。

0. 结论摘要

Strawmap 不是一份“官方承诺排期”，而是 EF Architecture 给 Ethereum L1 未来数年升级路线做的一张协调草图。它把升级放在三层里理解：共识层负责更快确认、更强抗审查、更轻/更后量子安全的验证者机制；数据层负责把 blob/DA 从 4844 时代推向 PeerDAS、blob streaming 和更大规模 DAS；执行层负责把 L1 从“每个节点重执行一切”的模式，逐步推向更高 gas、更清晰状态访问、更强证明、更规整交易/账户模型，以及未来可能的 native rollups 和 RISC-V/zkVM 路线。[S1][S2]

对 Mantle 来说，核心信号不是“L2 被 L1 吞掉”，而是“L1 会越来越像高吞吐 rollup settlement/DA/proof coordination layer”。最相关的变化是：blob 供给与定价、PeerDAS/后续 DA 扩容、L1 finality 缩短、native rollups/zkEVM 证明接口、状态访问与证明成本、交易/mempool 隐私。Mantle 的 EigenDA-first 与 SP1/ZK validity 方向仍然成立，但需要持续重估 Ethereum blob DA fallback、证明延迟、finality 语义和未来 native-rollup 兼容性。[S10][S31][S32][S45]

最重要的阅读方式：Strawmap 右侧的 fast L1、gigagas L1、teragas L2、post quantum L1、private L1 是 north stars，不是已验证指标或确定交付目标。当前更可靠的“近期 fork 状态”应以 EIP meta / Forkcast 语言表达：例如 ePBS/EIP-7732 与 BALs/EIP-7928 scheduled for Glamsterdam, FOCIL/EIP-7805 declined for Glamsterdam but scheduled for Hegotá, Frame Tx/EIP-8141 considered for Hegotá。不要把 Strawmap 的年份、箭头和北极星数字写成确定承诺。[S1][S3][S4][S5]

1. 如何读 Strawmap

Strawmap 有三个横向层：

- CL / Consensus Layer : slots、finality、roles、validator aggregation、post-quantum consensus。
- DL / Data Layer : blob/DA throughput、sampling、bloppool、data pricing、payload availability。
- EL / Execution Layer : gas limit、state、proofs、EVM hardening、transactions、mempool、native rollups。

图中的箭头可以理解为两类关系：硬技术依赖，或者自然演进方向。报告中会把“明确依赖”和“推断依赖”分开写。[S1][S2]

状态语言：

- `scheduled for X`：进入对应 meta EIP 的 `scheduled bucket`。
- `considered/proposed for X`：处于对应 fork 的考虑/提议范围，不能写成会进 fork。
- `research / offchain / concept`：有研究帖、repo 或路线图概念，但不是确定 protocol feature。
- `north star`：长期目标，主要用于协调方向，不是当前吞吐或交付承诺。[S3][S4][S5]

2. Layer 级别分析

2.1 共识层：当前问题与路线动机

共识层的瓶颈不是单一的“finality 慢”，而是一组被 12s slot、epoch finality、大验证者集合、PBS/relay、抗审查与未来 PQ 签名共同放大的限制。当前 L1 用户、bridge、交易所与 L2 都受慢 finality 影响；同时，验证者数量接近百万级时，全量 attestation 聚合与单槽 finality 会在网络带宽、聚合轮次、签名大小上遇到上限。[S6][S39][S45][S57][S58]

因此 CL 路线分成三条线：

- 延迟线：fast confirmation、quick slots、slot duration decreases、1-round finality。
- 角色线：ePBS、FOCIL、attester/includer separation、distributed block building。

- 安全/可持续线：validator set/issuance、lean consensus、PQ heartbeat、PQ pubkey registry、XMSS attestations、secret proposers。

2.2 数据层：当前问题与路线动机

L2 扩容的短中期瓶颈仍是 DA。EIP-4844 引入 blobs 后，blob gas 与 execution gas 分离，但当前 blob 供给、blobpool 传播、KZG 验证、fee floor 与低需求阶段的价格信号都还不是终局设计。PeerDAS 的目标是让节点不下载全部 blob 数据也能参与 DA 安全；后续 sparse blobpool、cell-level deltas、blob streaming 则是在降低传播和 critical path 压力。[S10][S11][S13][S14][S15]

因此 DL 路线从“更多 blobs”升级到“更高效、可抽样、可预传播、可定价的数据可用性”。teragas L2 / 1 Gbyte/sec 是 north star：它表达的是通过 DAS 支撑数量级更高 L2 DA，而不是当前 spec 已经给出可执行参数。[S1][S10]

2.3 执行层：当前问题与路线动机

执行层的瓶颈更复杂：单纯提高 gas limit 会把执行、状态读写、block propagation、witness、state root computation、precompile 特例、mempool validation 和 node hardware 压力一起放大。因此 Strawmap 的 EL 路线不是“把 gas limit 拉满”，而是先 repricing、BALs、state/proof plumbing、EVM hardening，再走 zkEVM/native rollup/mandatory proof 和可能的 RISC-V/LeanVM 路线。[S18][S19][S20][S24][S26][S27][S32][S43]

对 Mantle 最关键的是：Ethereum L1 如果逐步把执行验证转向 proof-heavy 模式，rollup 证明、状态访问和 native-rollup 兼容性会变成 L2 架构设计的一部分，而不只是链下运营优化。[S31][S32][S33][S34]

3. 路线图依赖关系

flowchart LR

A["EIP-4844 blobs"] --> B["PeerDAS / EIP-7594"]

B --> C["BP0 forks / blob capacity increments"]

B --> D["Sparse blobpool / cell-level deltas"]

D --> E["Blob streaming / ticketed AOT propagation"]

E --> F["Teragas L2 north star"]

G["ePBS / EIP-7732"] --> H["FOCIL / inclusion lists"]

H --> I["Attester-includer separation"]

J["Fast confirmation"] --> K["Quick slots"]

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K --> L["1-round / decoupled finality"]
L --> M["Fast L1 north star"]

N["BALs / EIP-7928"] --> O["Data/state repricing"]
O --> P["Gas-limit increases"]
P --> Q["Proof-heavy execution"]
R["EVMification / EIP-8200"] --> Q
Q --> S["Native rollups / EIP-8079"]
Q --> T["Block-in-Blobs / EIP-8142"]
T --> U["Gigagas L1 north star"]

V["Frame tx / EIP-8141"] --> W["Keyed nonces + recent roots"]
W --> X["Encrypted/privacy mempool"]
X --> Y["Private L1 north star"]
Z["PQ key registry"] --> ZA["PQ heartbeat / XMSS attestations"]
ZA --> ZB["Post-quantum L1 north star"]
```

明确依赖里，Fusaka/PeerDAS/BPO、Glamsterdam/ePBS/BALs、Hegotá/FOCIL 的状态最清晰。长程 I/J/K/L 主要是 Strawmap/PQ roadmap 里的规划链条，应以 milestone 或 research direction 表达。[S3][S4][S12][S46]

3.1 Strawmap 标签覆盖表

下表把 Google Drawing/PDF 中提取出的路线图标签映射到本报告中的分析位置。
merged 表示它不是单独 EIP，而是并入相邻机制讨论；unresolved 表示没有找到足够明确的 canonical source，只保留为 Strawmap concept。

Strawmap 标签	覆盖方式	主要来源
fast L1 / finality in seconds / fast confirmation	Fast confirmation / FCR 行	[S1][S6]
slot duration decreases / quick slots	Slot duration decreases 与 Quick slots 行	[S3][S5][S7]
SFI	SFI 行，作为 EIP-7723 process status	[S5]
ePBS / attester-proposer separation / distributed block building	ePBS、Attester-proposer / includer separation、Distributed block building 行	[S8][S56]
FOCIL	FOCIL 行	[S3][S4][S9]
decoupled consensus / 1-round finality	Decoupled consensus / 1-round finality 行	[S57][S44][S45]

Strawmap 标签	覆盖方式	主要来源
1M attestations per slot	1M attestations per slot 行	[S39][S58] [S59]
snail issuance	Snail issuance 行; exact phrase unresolved, 按 issuance/validator-set policy 映射	[S48][S49]
beacon & lean specs merge / tech debt reset / specs quantum	Beacon & lean specs merge / tech debt reset 与 PQ rows 合并覆盖	[S7][S40] [S41][S46]
51% attack auto-recovery	新增 51% attack auto-recovery 行	[S50]
post quantum heartbeat / post quantum pubkey registry / PQ leanXMSS attestations	对应 PQ rows	[S38][S39] [S41]
VDF randomness / real-time CL proofs / secret proposers	对应 CL rows	[S51][S40] [S41][S52] [S53]
teragas L2 / 1 Gbyte/sec / data availability increases	DA increases / PeerDAS 与 north-star caveat	[S1][S10]
sparse blobpool / cell-level deltas / local blob reconstruction / proofs of custody	DL rows	[S10][S13] [S14]
blob streaming / short-dated blob futures	DL rows ; short-dated blob futures 映射到 EIP-8256/ticketed reservation 而非金融 futures	[S15]
remove blob transaction type / leanVM / PQ leanDA sampling / leanDA block sampling / partial binary tree	DL/EL rows ; leanDA exact label unresolved	[S31][S34] [S40][S41] [S46]
data repricing / multidimensional pricing	DL rows	[S17][S18] [S21][S22] [S23]
gigagas L1 / 1 Ggas/sec / gas limit increases	EL rows and north-star caveat	[S1][S18] [S19]
ethp2p broadcast / ethp2p unification / sharded mempool	ethp2p broadcast/unification 与新增 Sharded mempool 行	[S25][S37]
evm-asm canonical guest / state-asm / zkzkRISC-V frames	EL proving/ISA rows	[S26][S43]
Glamsterdam repricing / BALs	EL rows	[S20][S47]
optional 2-of-3 proofs / mandatory 1-of-1 proofs	Proof rows ; 数字短语按 Strawmap interpretation 处理, 不作为 canonical term	[S24][S32]
validity-only partial state / decentralized state / endgame state / pureth purges	State rows	[S33][S34] [S35]
EVMify long-tail precompiles	EIP-8200 row	[S27]

Strawmap 标签	覆盖方式	主要来源
frame transactions / keyed nonces & recent roots / ephemeral keys	Transaction/account rows	[S28][S29] [S30][S42]
native rollups	Native rollups row	[S31]
long-dated gas futures	Long-dated gas futures row; research only, no canonical EIP	[S54][S55]
PQ leanSPHINCS transactions	新增 row ; mapped to frame/PQ account-auth track, exact Strawmap label not a canonical EIP	[S28][S42] [S46]
privacy mempool / encrypted mempool / private L1 / lean privacy wormholes	Privacy rows; lean privacy wormholes unresolved and mapped to privacy/PQ/leanVM research track	[S1][S36] [S41][S42]

4. 优化点明细

4.1 共识层优化点

优化点	当前瓶颈	原理/机制	引入后的变化	状态/依赖
Fast confirmation / FCR	Ethereum finality 对 UX、bridge、交易所 有约 15 min 级等待压力	consensus-client 规则基于 attestation 条件给出一槽级确认，失败时回退到 finalized head	正常网络下可给出约 13s 级 confirmation，但安全性弱于 finality	client-side/offchain；不需要 hard fork [S6]
Slot duration decreases	12s slot 是 L1 inclusion latency 的基础下限	缩短 slot，比如 12s -> 6s，需要重新分配 proposal/attestation/aggregation deadline	inclusion 与 finality cadence 加快，但传播/验证压力上升	EIP-7782 在相关讨论中出现，但 Glamsterdam meta 当前将其列为 declined；需谨慎 [S3][S5]
Quick slots / EIP-8198	CL 客户端到处硬编码 12s 假设	先把 slot duration runtime-configurable，再 benchmark、逐步缩短	未来 slot reduction 从大改变成可配置路径	Draft；更像基础设施准备 [S7]
SFI	fork scope 容易把“想研究”和“准备进入 fork”混淆	EIP-7723 定义 Scheduled for Inclusion 等状态	fork 治理语言更清晰	process status，不是协议功能 [S5]
ePBS / EIP-7732	PBS 依赖 relay/off-protocol trust，执行 payload 在 critical path	enshrine proposer-builder separation，用 commitment/PTC/延迟执行验证	降低 relay 信任和 critical-path payload 压力，改善 builder/proposer 角色分离	Draft；scheduled for Glamsterdam；free-option/复杂性仍是风险 [S3][S8]

优化点	当前瓶颈	原理/机制	引入后的变化	状态/依赖
FOCIL / EIP-7805	builder 集中使抗审查弱化	committee 生成 inclusion lists, fork choice 强制满足 IL	提升 forced inclusion 与 censorship resistance	Draft ; declined for Glamsterdam, scheduled for Hegotá [S3][S4][S9]
Attester-proposer / includer separation	inclusion 与 state execution 混在一起, 激励和费用不清晰	把 inclusion tx 与 state tx、includer 与 proposer/builder 角色拆开	更清晰的 inclusion fee 与抗审查路径	Research ; 逻辑上在 PBS/FOCIL 之后 [S56]
Distributed block building	builder/relay 集中形成 chokepoint	多方或分布式 builder/proposer 协作	降低单点与 relay 依赖, 但协调复杂度上升	Research / PBS 相关 [S8]
Decoupled consensus / 1-round finality	单槽 finality 与全量验证者投票/聚合存在 tradeoff	heartbeat chain 与 trailing finality gadget 解耦	有机会在不让每个 slot 承担全量 finality 成本的情况下加快 finality	Research ; 依赖 committee、aggregation、networking [S57][S44][S45]
1M attestations per slot	大验证者集合下 naive 全量投票不可扩展	小 heartbeat committee + 后置 finality / aggregation	让 fast path 更轻, 同时保留更强 finality	Research ; 数字是压力场景, 不是 spec 参数 [S39][S58][S59]
Snail issuance	验证者集合增长会放大 CL 通信与聚合成本	exact phrase 未找到 canonical source ; 最接近 Orbit SSF / issuance policy 对 validator-set growth 与 issuance curve 的讨论	可能降低长期 CL scaling 压力, 但经济安全、solo staking 与中心化权衡未定	unresolved label ; 按 issuance/validator-set policy research 处理 [S48][S49]
Beacon & lean specs merge / tech debt reset	beacon specs 与 lean/PQ/证明方向复杂度累积	lean consensus/spec tooling 与 runtime-configurable timing	降低客户端/规格技术债, 为 PQ/fast finality 做准备	offchain/spec work ; EIP-8198/leanSpec 相关 [S7][S40][S41]
Post-quantum pubkey registry	validator PQ migration 需要提前生成/登记密钥, 不能等激活时突击	先登记 PQ public keys, 为 XMSS/哈希签名迁移做 warmup	降低迁移操作风险	Research ; 不是 finalized design [S38]
Post-quantum heartbeat	全量 PQ aggregation 太重	小 heartbeat committee 可直接拼接 PQ signatures, finality 另走压缩/证明路径	让 PQ liveness fast path 更现实	Research ; 依赖 dynamic availability/leanMultisig [S39][S41]
PQ leanXMSS attestations	BLS/ECDSA 长期受量子威胁; XMSS stateful 且大签名	leanMultisig/leanVM 用递归证明压缩 XMSS 签名集合	提供 CL PQ attestation 方向, 但 signer state discipline 很重	Implementation/research ; repo benchmark 不是协议保证 [S41]

优化点	当前瓶颈	原理/机制	引入后的变化	状态/依赖
VDF randomness	committee/ proposer randomness 需要抗操纵	RANDAO+VDF 等历史路线	提升 randomness 质量，但与新设计优先级关系不明	历史研究，当前 Strawmap 中属长程概念 [S51]
Real-time CL proofs	consensus 验证和 light-client/证明仍不够实时	ZK-verified consensus / leanVM / recursive proofs	未来可让 CL 验证更轻、更可组合	Aspirational research [S40][S41]
Secret proposers	proposer 身份提前暴露可被 DoS	Whisk/SSLE 等秘密领导者选择	提升 proposer privacy 和 DoS resistance	Research；非近期承诺 [S52][S53]
51% attack auto-recovery	严重多数攻击后，在线节点需要一致判断哪个分叉/区块是 timely 或应恢复的链	timeliness detectors 让客户端在同步假设下判断区块是否准时，并帮助识别/恢复攻击后的 canonical view	目标是降低攻击后长期混乱与社会恢复负担，但依赖网络延迟假设和在线多数	Research；不是当前 fork commitment [S50]

4.2 数据层优化点

优化点	当前瓶颈	原理/机制	引入后的变化	状态/依赖
DA increases / PeerDAS	blob 扩容如果要求节点全量下载，会推高带宽/存储	PeerDAS 用 erasure coding、columns、sampling 让节点下载一部分数据	增加 blob/DA capacity 同时不线性增加节点负担	EIP-7594 是核心 DA 升级；Fusaka/Fulu 相关 [S10]
Blob target/max bump	4844 后 blob supply 有短期供给约束	EIP-7691 提升 target/ max blobs	先给 L2 DA 更多 headroom	近中期容量 bump [S11]
BPO forks	每次调 blob 参数都 full hard fork 成本高	EIP-7892 允许 blob-parameter-only forks	blob capacity 可更频繁调整	PeerDAS/BPO 扩容工具 [S12]
Sparse blobpool / EIP-8070	full-replication blobpool 占用 EL 带宽并影响 liveness	provider/sampler 角色分离，概率性 full fetch + custody-aligned sampling	降低 blobpool 平均带宽，保留可用性验证路径	Draft networking EIP [S13]
Cell-level deltas / EIP-8136	节点已有大部分 cells 时仍重复交换整列/大量数据	用 bitmap/request-response 只交换未见 cells	降低 PeerDAS churn 与冗余带宽	Draft；PeerDAS optimization [S14]
Local blob reconstruction	DAS 下并非所有节点都有完整 blobs	基于采样/cell/proof 重建所需 blob data	增强本地验证与恢复能力	与 PeerDAS/ sparse blobpool 相关 [S10][S13]
Proofs of custody	节点采样后需证明/约束自己 custody 的数据	custody sampling + proofs / subnet responsibilities	使 DAS 安全不依赖全量下载	PeerDAS/DAS family [S10] [S13]

优化点	当前瓶颈	原理/机制	引入后的变化	状态/依赖
Blob streaming / ticketed AOT	blob data propagation 仍在 critical path, naive EL pre-propagation 有 DoS 风险	ticket-based ahead-of-time blobs + JIT blobs	将部分 blob propagation 移出 critical path, 给 L2 sequencer 更稳定提交窗口	Draft/HackMD ; EIP-8256 closest match [S15]
Short-dated blob futures	L2 需要提前锁定/平滑 blob capacity/cost	Strawmap 语境最接近 EIP-8256 ticketed reservation, 而非金融 futures	更像容量预留/传播设计, 不是 derivatives market	概念/研究; 谨慎命名 [S15]
Remove blob transaction type	未来 native-rollup/proof-heavy 模式下, blob tx 与 execution semantics 可能需规整	EIP-8079/native rollups 不支持 blob tx, 或将 DA 与 execution 接口重新组织	简化 native rollup STF 与 execution/DA 边界	Draft/native rollup 路线 [S31]
Data repricing	blob/call/BAL bytes 的真实资源成本不匹配	EIP-8131/8279/8311 等给 byte-dense 资源加 floor	避免扩容后被便宜字节拖垮传播/状态	Proposed/repricing cluster [S18] [S21][S22] [S23]
Multidimensional pricing / EIP-7999	单一 gas 无法同时给 execution、data、state、blob 正确定价	多资源维度统一 max_fee/reserve pricing	用户预算更统一, 资源定价更真实, 但 builder packing 更复杂	Draft ; 结构性 fee market 改造 [S17]
PQ leanDA sampling / leanDA block sampling	长期 DA 还要考虑 PQ commitment 与更大规模 sampling	Hash/PQ-friendly DA commitments + sampling	支撑 post-quantum L2 DA north star	没有找到精确 canonical label ; 按 lean/PQ DAS research 处理 [S40][S46]
Partial binary tree	state/proof/witness 需要 PQ-friendly 树结构	binary tree/hash-based state commitments	减少 witness 并避免 Verkle 曲线长期 PQ 风险	更接近 EL/state ; DL 图上作为数据结构 hardening [S34]

4.3 执行层优化点

优化点	当前瓶颈	原理/机制	引入后的变化	状态/依赖
BALs / EIP-7928	state access 隐含在执行中, 难以并行/预取/验证	block-level access lists 显式记录访问账户/slot 与结果	支持并行 disk reads、validation、state-root computation、healing	Draft ; scheduled for Glamsterdam [S3][S20]

优化点	当前瓶颈	原理/机制	引入后的变化	状态/依赖
Glamsterdam repricing / EIP-8007	EVM gas 成本与实际瓶颈不匹配	作为 gas repricing directory 组织多个 repricing EIPs	为更高 gas 与 parallelism 清理资源错价	Meta/draft [S47]
Gas limit increases / EIP-7938	block capacity 受 gas limit 与 client coordination 限制	client default voting schedule 指数提升 gas limit	提供容量提升路径, 但 state growth/ propagation 风险高	Draft/非共识默认策略; 不是直接承诺 [S19]
EIP-8131/8279/8311 repricing	calldata、BAL bytes、byte-dense cases 可绕过 floor	64/96 gas/byte 等 floors + BAL runtime metering	先收紧最坏块大小, 再考虑 450M gas 级别分析目标	Proposed/analysis; 不是已定参数 [S18][S21][S22][S23]
ethp2p broadcast/unification	generic p2p 对大对象传播低效; CL/EL 带宽缺少统一控制	erasure-coded broadcast、object-specific strategy、QUIC/ECH、slot-aware control	改善 blocks/ blobs/BAL/ PTC/IL 等传播效率	Repo/spec research; broadcast 已较具体, 其它层仍研究 [S25]
Optional execution proofs / EIP-8025	beacon nodes 依赖 EL re-execution 验证 payload	CL 可选择使用 execution proofs	降低每个 verifier 跑 EL 的必要性	Stagnant; 作为 proof path concept [S24]
Mandatory proofs / 1-of-1 proofs	gigagas L1 下所有节点重执行不可持续	zkEVM/real-time proofs 让验证转向 proof verification	execution validation 更轻, 但 proving/state availability 成关键瓶颈	Strawmap concept; 具体由 EIP-8142/native rollups/ zkEVM stack 支撑 [S24][S32]
Block-in-Blobs / EIP-8142	如果 validators 不再重执行, execution payload data availability 不能靠 re-execution 隐含保证	把 payload data 放入 blobs, 并在 EL/CL 间绑定 blob commitments	proof-heavy execution 仍保持 DA	Draft; 依赖 4844/7594/7732/7892/7928 [S32]
Native rollups / EIP-8079	rollups 仍依赖外部证明/桥接信任模型	L1 暴露/验证 Ethereum STF execution primitive	rollups 更接近继承 L1 security 与同步组合性	Draft; 长期架构方向 [S31]

优化点	当前瓶颈	原理/机制	引入后的变化	状态/依赖
evm-asm canonical guest	zkEVM guest 与 compiler trust/spec conformance 复杂	Lean 4 verified macro-assembler over RV64 guest	提高 STF guest 可验证性, 减少 compiler trust	Prototype repo ; 非 production guest [S26]
state-asm / zkzkRISC-V frames	block execution proving 与 state/witness 成本高	verified state transition / RISC-V zkVM substrate	把 EVM execution 转成更适合证明的底层 ISA	Research/prototype ; RISC-V replacement 仍讨论 [S26] [S43]
LeanVM	PQ/lean stack 需要小型 zkVM/proof substrate	minimal hash-based zkVM + leanSig/compiler path	为 lean/PQ signatures/proofs 提供 VM-native 路线	Research repo [S41]
EVMify long-tail precompiles / EIP-8200	precompiles 是 client/zkEVM special cases	在相同地址部署等价 EVM bytecode, 停止 native precompile 处理	降低共识特例与 zkEVM 特例, 但 gas/perf 可能变化	Draft ; gas-sensitive contracts 风险 [S27]
Validity-only partial state / VOPS	weak statelessness 可能破坏 public mempool validity/censorship resistance	保留足够验证 pending tx 的部分状态	在降低存储的同时保留 mempool 验证能力	Research [S33]
Hyper-scaling state / new state forms	state 没有像 DAS/ZK 那样的单一 magic bullet	tiered/new state classes、temporary/UTXO-like storage	把 future state 设计成更便宜但更受限的资源	Research [S34]
Pureth purges / EIP-7919	RPC/blocks/receipts/tx data 可验证性不足	Pureth meta 关注 verifiable data access/RPC correctness	降低轻客户端/RPC 信任, 但不是直接 state-growth fix	Stagnant meta ; label 映射有不确定性 [S35]
Decentralized/endgame state	全状态存储、old state、witness 仍中心化	statelessness、expiry、state providers、new state forms	节点负担下降, 但 RPC/indexer/old-state retrieval 更重要	Research ; Verkle vs binary trie 仍有分歧 [S33][S34]
Frame transactions / EIP-8141	EOA/AA/PQ/key rotation/gas	transaction type 0x06 , frames for	原生 batching、gas sponsorship、	Draft ; considered for Hegotá [S4][S28]

优化点	当前瓶颈	原理/机制	引入后的变化	状态/依赖
	sponsorship 被现有 tx 格式限制	validation/ payment/ execution	key rotation、 PQ-friendly auth	
Keyed nonces / EIP-8250	shared sender/ global nonce 串行化隐私协议和 relayers	(nonce_key, nonce_seq) 拆分 nonce domains	提高同 sender 并发与 privacy/ nullifier workflows	Draft/proposed for Hegotá [S4][S29]
Recent roots / EIP-8272	validation 需要引用近期 roots, 但直接读 mutable state 易失效/DoS	frame tx 引用 (source_id, slot, root), 预检查后暴露给 validation code	privacy trees、auth roots、wallet roots 更易安全绑定	Draft/proposed [S30]
Ephemeral keys	长期 ECDSA public-key exposure 有 PQ 风险	AA/smart wallet 每笔交易/短周期换 key, 或 hash-based few-time signatures	降低长期 key exposure, 但 pending mempool window 仍暴露	Wallet/AA research ; 需 private mempool 缓解 [S42]
Announce tx with nonce / EIP-8077	mempool propagation 缺少 sender/nonce hints	tx announcements 带 source address 和 nonce	支持更精细 mempool/sharding/validation	Draft networking [S37]
Sharded mempool	单一公共 mempool 在 blob/tx/nonce/privacy 路线下传播与验证压力上升	目前没有单一 canonical EIP ; 最接近的是 sparse blobpool、announce-with-nonce、encrypted/private mempool 的组合	可降低传播压力并让不同对象走专门路径, 但会增加 mempool policy 与 DoS 面	merged concept ; 见 EIP-8070/EIP-8077/EIP-8184 [S13][S37][S36]
Encrypted / privacy mempool / EIP-8184	public mempool 暴露内容和排序机会	sealed tx/bundle、commit-before-reveal、delayed key release	降低 front-running/censorship/order-flow 泄露, 但引入 reveal optionality/DoS/key trust	Draft ; depends on FOCIL [S36]
Private L1 / shielded transfers	L1 原生隐私弱, 应用层隐私依赖外部协议	shielded transfer、privacy mempool、PQ-	隐私成为 L1 一等能力, 但 UX/compliance/ciphertext bloat 很难	North star / research [S1] [S36][S42]

优化点	当前瓶颈	原理/机制	引入后的变化	状态/依赖
		safe note discovery		
PQ leanSPHINCS transactions	账户签名长期需要 PQ-safe 替代, 且现有 EOA/tx envelope 不适合大 PQ 公钥/签名	exact label 无 canonical EIP ; 最接近 frame transactions + PQ account-auth / hash-based signature research	可作为 post-quantum account-auth path, 但签名大小、mempool、alias/key registry 仍未定	Strawmap concept ; mapped to frame/PQ AA track [S28][S42][S46]
Lean privacy wormholes	Strawmap 标签指向 lean/PQ/privacy 交叉方向, 但没有找到同名 canonical source	暂按 leanVM + encrypted/privacy mempool + shielded-transfer research 合并处理	表示长期 privacy primitive / proof-system 方向, 不应写成确定功能	unresolved label ; merged into privacy/PQ research track [S36][S41][S42]
Long-dated gas futures	L2/应用需要长期 gas 成本确定性	研究中的 in-protocol gas futures/secondary markets	可做 treasury/user fee hedging, 但机制未定	Research ; 非 canonical EIP [S54][S55]

5. Mantle dev team 重点影响

1. DA strategy 不应静态化。即使 Mantle 当前继续 EigenDA-first, Ethereum blobs/PeerDAS/BPO/streaming 会改变 Ethereum fallback DA 的成本、可用性和市场风险。Mantle 应持续维护三套模型: EigenDA-only、Ethereum blob fallback、mixed DA。[S10][S11][S12][S15]
2. Blob pricing 会从“便宜 DA”走向更复杂的资源市场。EIP-7918/EIP-7999/8131/8279/8311 说明 Ethereum 正在处理 blob 与 execution/basefee 之间的价格信号失真。Mantle 若提供稳定 fee 产品, 需要把 blob congestion 与 reserve pricing 纳入 treasury/ops 模型。[S16][S17][S18][S21][S22]
3. Faster L1 finality 会压缩 bridge 和 settlement UX 预期, 但不会自动消除 L2 proof/finality 延迟。Mantle 对用户说明 finality 时, 需要区分 L1 confirmation/finality、L2 validity proof finality、withdrawal finality。[S6][S45]

4. Native rollups 与 proof-heavy L1 会改变 “Ethereum compatibility” 的含义。未来兼容性可能不只是 EVM bytecode, 还包括 proof interface、state/witness format、recent-root/auth-root/mempool rules。[S31][S32][S33]
5. Execution/state 路线会影响 prover operations。VOPS、BALs、state forms、BiB、evm-asm/RISC-V 都指向同一个事实：证明成本和状态访问模式会成为 L2 性能边界。Mantle 的 SP1/prover roadmap 应持续关注 Ethereum L1 对 witness、state root、payload DA 的标准化方向。[S20][S26][S32][S33][S34]
6. Mempool/privacy 预期会上升。若 L1 推进 FOCIL、LUCID、privacy mempool、shielded transfers, 用户也会更关注 L2 sequencer ordering、fair ordering、private order flow。Mantle 应明确自己是否要 mirror L1 privacy/censorship-resistance features。[S9][S36][S37]

6. 主要不确定性与风险

- Strawmap 是 draft, 不是 commitment。七个 forks 到 2029、六个月 cadence、north star 数字都应写成协调假设。[S1]
- EIP 状态变化快。报告中的 fork bucket 以 2026-07-06 抓取的 EIP meta/Forkcast 为准；后续需要复查。[S3][S4][S5]
- FOCIL 状态容易误读。它不在 Glamsterdam scheduled list, 而是 Hegotá scheduled；旧讨论仍可能把它作为 Glamsterdam 候选，需要注明时间差。[S3][S4][S9]
- 6s slots / EIP-7782 在不同页面/讨论中可能出现漂移；当前不应写成确定 Glamsterdam feature。[S3][S5]
- snail issuance、leanDA、short-dated blob futures、long-dated gas futures、lean privacy wormholes 等标签没有全部找到精确 canonical source；报告中应标为 inferred/concept。
- 数字 claim 要降级：1 Ggas/s、1 GB/s、10K TPS、10M TPS 是 north star；450M/600M gas、PeerDAS sampling fraction、bandwidth reduction 是 proposal/model claim, 不是交付承诺。[S1][S10][S18][S19]

7. 建议给 Mantle 的行动清单

1. 建立 DA cost sensitivity dashboard：current blobs、PeerDAS/BPO-expanded blobs、EigenDA-only、mixed fallback 四种情景。
2. 把 “finality” 文档拆成三层：L1 confirmation/finality、L2 proof finality、bridge withdrawal finality。

3. 对 native rollup readiness 做架构评估：是否需要兼容未来 EXECUTE /L1-verified STF 类接口。
4. 给 prover team 建立 Ethereum L1 proof/state watchlist : BALs、VOPS、BiB、EIP-8141/8272、evm-asm/RISC-V、state tree alternatives。
5. 对 sequencing/mempool 做隐私与抗审查差距分析：FOCIL、encrypted mempool、sender/nonce leakage、private order flow。
6. 在研究报告或内部分享中，避免把 Strawmap north stars 写成“EF 确认的交付目标”。

Sources

- [S1] Strawmap landing page and FAQ, <https://strawmap.org/>, local HTTP 200 export, accessed 2026-07-06.
- [S2] Strawmap Google Drawing export, https://docs.google.com/drawings/d/1GkcGfQv9kxgrYQMyu0BYGcZya0gIDG_wQ7GsFJEsdzY/export/png, accessed 2026-07-06.
- [S3] EIP-7773 Hardfork Meta - Glamsterdam, <https://eips.ethereum.org/EIPS/eip-7773>.
- [S4] EIP-8081 Hardfork Meta - Hegotá, <https://eips.ethereum.org/EIPS/eip-8081>.
- [S5] EIP-7723 process/status definitions, <https://eips.ethereum.org/EIPS/eip-7723>.
- [S6] Fast Confirmation Rule, <https://fastconfirm.it>.
- [S7] EIP-8198 Quick Slots, <https://eips.ethereum.org/EIPS/eip-8198>.
- [S8] EIP-7732 Enshrined Proposer-Builder Separation, <https://eips.ethereum.org/EIPS/eip-7732>.
- [S9] EIP-7805 FOCIL, <https://eips.ethereum.org/EIPS/eip-7805>.
- [S10] EIP-7594 PeerDAS, <https://eips.ethereum.org/EIPS/eip-7594>.
- [S11] EIP-7691 Blob throughput increase, <https://eips.ethereum.org/EIPS/eip-7691>.
- [S12] EIP-7892 Blob-parameter-only forks, <https://eips.ethereum.org/EIPS/eip-7892>.
- [S13] EIP-8070 Sparse Blobpool, <https://eips.ethereum.org/EIPS/eip-8070>.
- [S14] EIP-8136 Cell-Level Deltas, <https://eip.tools/eip/8136>.
- [S15] Blob streaming HackMD / EIP-Blob-streaming, https://hackmd.io/@gMH0wL_0Sr69C7I7in08MQ/B125sCUtZx.

- [S16] EIP-7918 Blob base fee bounded by execution cost, <https://eips.ethereum.org/EIPS/eip-7918>.
- [S17] EIP-7999 Unified multidimensional fee market, <https://eips.ethereum.org/EIPS/eip-7999>.
- [S18] Hegotá Repricings, <https://misilva73.github.io/hegota-repricings/>.
- [S19] EIP-7938 Exponential Gas Limit Increase, <https://eips.ethereum.org/EIPS/eip-7938>.
- [S20] EIP-7928 Block-Level Access Lists, <https://eips.ethereum.org/EIPS/eip-7928>.
- [S21] EIP-8131 transaction-content byte floor, <https://raw.githubusercontent.com/ethereum/EIPs/master/EIPS/eip-8131.md>.
- [S22] EIP-8279 BAL byte floor, <https://raw.githubusercontent.com/ethereum/EIPs/master/EIPS/eip-8279.md>.
- [S23] EIP-8311 calldata repricing, <https://raw.githubusercontent.com/ethereum/EIPs/master/EIPS/eip-8311.md>.
- [S24] EIP-8025 Optional Execution Proofs, <https://eips.ethereum.org/EIPS/eip-8025>.
- [S25] ethp2p repo, HEAD 741d8d9cf682ff93b7d8eb56e0377ba8eea83a7e, <https://github.com/ethp2p/ethp2p>.
- [S26] Verified-zkEVM/evm-asm repo, HEAD 864fbe8ced0dee1bf5d4371b72d025a5a9319f97, <https://github.com/Verified-zkEVM/evm-asm>.
- [S27] EIP-8200 EVMification, <https://eips.ethereum.org/EIPS/eip-8200>.
- [S28] EIP-8141 Frame Transaction, <https://eips.ethereum.org/EIPS/eip-8141>.
- [S29] EIP-8250 Keyed Nonces for Frame Transactions, <https://eips.ethereum.org/EIPS/eip-8250>.
- [S30] EIP-8272 Recent Roots for Frame Transactions, <https://eips.ethereum.org/EIPS/eip-8272>.
- [S31] EIP-8079 Native rollups, <https://eips.ethereum.org/EIPS/eip-8079>.
- [S32] EIP-8142 Block-in-Blobs, <https://eips.ethereum.org/EIPS/eip-8142>.

- [S33] VOPS research, <https://ethresear.ch/t/a-pragmatic-path-towards-validity-only-partial-statelessness-vops/22236>.
- [S34] Hyper-scaling state research, <https://ethresear.ch/t/hyper-scaling-state-by-creating-new-forms-of-state/24052>.
- [S35] EIP-7919 Pureth Meta, <https://eips.ethereum.org/EIPS/eip-7919>.
- [S36] EIP-8184 LUCID encrypted mempool, <https://eips.ethereum.org/EIPS/eip-8184>.
- [S37] EIP-8077 announce transactions with nonce, <https://eips.ethereum.org/EIPS/eip-8077>.
- [S38] PQ public-key registry research, <https://ethresear.ch/t/exploring-the-design-space-for-a-post-quantum-public-key-registry-for-ethereum-validators/25040>.
- [S39] Dynamic availability / PQ heartbeat research, <https://ethresear.ch/t/why-ethereum-needs-a-dynamically-available-protocol/24418>.
- [S40] Lean Ethereum EF post, <https://blog.ethereum.org/2025/07/31/lean-ethereum>.
- [S41] leanEthereum/leanMultisig / leanVM repo, HEAD 12e61512416548e743040aab4daf83c58a5c5476, <https://github.com/leanEthereum/leanMultisig>.
- [S42] Ephemeral keys and account abstraction research, <https://ethresear.ch/t/achieving-quantum-safety-through-ephemeral-key-pairs-and-account-abstraction/24273>.
- [S43] RISC-V execution-layer proposal, <https://ethereum-magicians.org/t/long-term-l1-execution-layer-proposal-replace-the-evm-with-risc-v/23617>.
- [S44] One-round finality note, <https://notes.ethereum.org/@yannvon/S1wIXIDqbe>.
- [S45] ethereum.org single-slot finality page, <https://ethereum.org/roadmap/single-slot-finality/>.
- [S46] ethereum.org quantum-resistance roadmap, <https://ethereum.org/roadmap/future-proofing/quantum-resistance/>.
- [S47] EIP-8007 Glamsterdam Gas Repricings, <https://eips.ethereum.org/EIPS/eip-8007>.
- [S48] Orbit SSF validator-set management research, <https://ethresear.ch/t/orbit-ssf-solo-staking-friendly-validator-set-management-for-ssf/19928>.

- [S49] Practical endgame on issuance policy, <https://ethresear.ch/t/practical-endgame-on-issuance-policy/20747>.
- [S50] Timeliness detectors and 51% attack recovery, <https://ethresear.ch/t/timeliness-detectors-and-51-attack-recovery-in-blockchains/6925>.
- [S51] Minimal VDF randomness beacon, <https://ethresear.ch/t/minimal-vdf-randomness-beacon/3566>.
- [S52] Whisk practical shuffle-based SSLE, <https://ethresear.ch/t/whisk-a-practical-shuffle-based-ssle-protocol-for-ethereum/11763>.
- [S53] Secret non-single leader election, <https://ethresear.ch/t/secret-non-single-leader-election/11789>.
- [S54] On in-protocol gas futures, <https://ethresear.ch/t/on-in-protocol-gas-futures/23698>.
- [S55] How to purchase Ethereum GAS in advance, <https://ethresear.ch/t/how-to-purchase-ethereum-gas-in-advance/19069>.
- [S56] Towards Attester-Includer Separation, <https://ethresear.ch/t/towards-attester-includer-separation/21306/1>.
- [S57] Unblocking faster finality with decoupled consensus, <https://ethresear.ch/t/unblocking-faster-finality-with-decoupled-consensus/24527/1>.
- [S58] Horn: collecting signatures for faster finality, <https://ethresear.ch/t/horn-collecting-signatures-for-faster-finality/14219>.
- [S59] LMD GHOST with 256 validators and a fast-following finality gadget, <https://ethresear.ch/t/lmd-ghost-with-256-validators-and-a-fast-following-finality-gadget/22856>.

Methodology Appendix

- First wave: 14 parallel research lanes covered Strawmap inventory, CL latency, CL/PQ scalability, DA throughput, DA economics, EL throughput, EL proving, EL state, EIP metadata, linked repos/docs, dependency graph, Mantle implications, PQ/privacy, skeptic audit.
- Expansion waves: fork-meta reconciliation; execution/privacy interfaces; gas/pricing/EVM hardening crosswalk.

- Local artifacts: `strawmap.html`, `strawmap.svg`, `strawmap.pdf`, `strawmap.png`, `strawmap-pdf.txt`, `strawmap-links-clean.txt`, `sources-fetch-index.json`.
- Access gap: linked Google published doc returned HTTP 401/404 in this environment; it is not used as a report source.